Project 2

TITLE: Automated Irrigation Controller

OBJECTIVE:

To design an automated irrigation system that monitors soil moisture and controls water supply automatically.

COMPONENTS USED:

* ESP32

* Potentiometer (Soil Moisture Sensor Simulation)

* LED (Water Pump Simulation)

* Wokwi Simulator

SOFTWARE USED:

* Arduino IDE

* Wokwi Simulator

WORKING PRINCIPLE:

The potentiometer simulates soil moisture levels. The ESP32 reads moisture values through ADC and decides whether irrigation is required. If moisture falls below the threshold, the simulated pump is activated.

FEATURES:

* Soil moisture monitoring

* Automatic irrigation control

* Moisture percentage calculation

* Dry/Wet soil classification

* Hysteresis control

RESULT:

The irrigation system successfully controlled the water pump based on soil moisture levels.

CONCLUSION:

This project demonstrated sensor-based automation and actuator control using ESP32.